Kavi Mookherjee Amodt

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Education

UC Berkeley — Expected August 2026

B.A. Computer Science and Cognitive Science; Linguistics Minor — *GPA: 3.72*

Relevant Coursework: Probability & Random Processes, Artificial Intelligence, Algorithms, Computer Security, Software Engineering, Algorithmic Economics, Computability & Complexity, Econometrics, Computer Architecture, Data Structures, Discrete Math & Probability, Linear Algebra, Statistics

Projects

Secure File Sharing System

- Designed and implemented a cryptographically secure file sharing system in Go using public-key encryption, digital signatures, and HMACs.
- Validated correctness under a fully adversarial server environment where attackers hold full write access to the datastore and read access to the keystore.

Regression and Gradient Descent From Scratch

- Implemented linear and logistic regression from first principles with gradient descent and closed-form solvers in Python/NumPy.
- Added ridge (L2) and lasso (L1) regularization and tuned penalty strength via k-fold cross-validation, using bias-variance analysis to diagnose over- and underfitting.

Assembly Level Digit Classifier

- Implemented a neural network inference engine in RISC-V assembly, capable of classifying handwritten digits through matrix operations and ReLU activation.

MCMC Cipher Decoder

- Implemented the Metropolis-Hastings algorithm from scratch in Python/NumPy to sample from arbitrary target distributions.
- Applied MCMC to decrypt substitution ciphers by modeling key permutations as a Markov chain.

Prediction Market Simulator

- Simulated and visualized prediction market dynamics using the LMSR online learning algorithm in C++.

Experience

Software Engineering Intern — May 2024 – Aug. 2024

Tools for Humanity — *San Francisco, CA*

- Developed production Rust, Python, and Bash code for an iris-scanning biometric device (the Orb).
- Engineered Rust CLI tooling to surface and query real-time Orb device metrics.

Research Assistant — Jan. 2025 – Present

Language and Cognitive Development Lab — *UC Berkeley*

- Designed and implemented computational experiment for word-object learning models using dynamic field theory in MATLAB.
- Developed automated data processing and analysis pipelines in Python.

Research Assistant — 2021 – 2023

UCSC Linguistics Department — *Santa Cruz, CA*

- Conducted eye-tracking research on syntactic ambiguity resolution and lexical underspecification.

Skills

Languages: Python, Rust, Go, C++, C, Java, R, RISC-V Assembly, SQL, Bash
Libraries & Frameworks: PyTorch, NumPy, SciPy, pandas, Matplotlib
Tools & Platforms: Git, Jupyter, Linux, MATLAB, Model Context Protocol (MCP)
Domains: Machine learning, deep learning, applied cryptography, information theory, probabilistic modeling, computer architecture